

13.1: Exponential Functions

Equations of form $y = a^x$ are exponential. a is the base, & x is the exponent

When $x > 0$ / $a > 1$: Exponential Growth
When $x < 0$ / $a < 1$: Exponential Decay.

$\log_a x = y$
$\Rightarrow x = a^y$
$\Rightarrow \ln x = y \ln a$
$\Rightarrow \frac{\ln x}{\ln a} = \log_a x$

13.2: Logarithms

A logarithm is the exponent to which a base is raised to, in order to get a certain number.

$y = a^x \iff \log_a y = x$

Base of logarithms are 10 unless stated otherwise.

Log Laws:

$\log(ab) = \log(a) + \log(b)$
$\log\left(\frac{a}{b}\right) = \log(a) - \log(b)$
$\log(a^n) = n\log(a)$
$\log\left(\frac{1}{a}\right) = -\log(a)$
$\log(\sqrt[n]{a}) = \frac{1}{n}\log a$
$\log_a a = 1$
$\log_a 1 = 0$

13.4: Natural Logarithm

$y = e^x \iff \log_e y = x$

$\log_e y \equiv \ln y \Rightarrow \ln y = x$

Laws of logs still apply

$\frac{d(\ln x)}{dx} = \frac{1}{x}$

13.3: Exponential Function.

For continuous interesting, e is used.
For compound interest: $A = A_0\left(1 + \frac{r}{n}\right)^{nt}$

Let $u = \frac{n}{r} \iff n = ur \Rightarrow \left[\left(1 + \frac{1}{u}\right)^u\right]^r$

$\lim_{n\to\infty} \Rightarrow \lim_{u\to\infty}\left(1 + \frac{1}{u}\right)^u = e$

$\Rightarrow A_0 e^{rt} = A$ For continuous Interest.

$\frac{d(e^x)}{dx} = e^x \quad \frac{d(e^{kx})}{dx} = ke^{kx}$

13.5: Modelling Curves:

It is easy to find a relationship between variables if they lie on a straight line ($y = mx + c$)

$y = ax^b \Rightarrow \ln y = \ln a + b \ln x$
$c = \ln a$, $m = b$, $\ln y$ VS. $\ln x$

$y = ab^x \Rightarrow \ln y = \ln a + x \ln b$
$c = \ln a$, $m = \ln b$, $\ln y$ VS. x